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journal homepage: www.journals.elsevier.com/journal-of-accounting-and-economicsDoes generative AI facilitate investor Trading? Early evidence from ChatGPT outages[☆]Qiang Cheng^{*} , Pengkai Lin[✉] , Yue Zhao[✉] 

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ABSTRACT

In this paper, we use ChatGPT outages to provide early evidence on whether investors rely on generative artificial intelligence (GenAI) to perform professional tasks and the associated impact on stock price informativeness. We document a significant decline in stock trading volume during ChatGPT outages. The effect is stronger for firms with corporate news released immediately before or during the outages and for firms with higher ownership held by transient institutional investors. We then document declines in short-run price impact and return variance during the outage periods, consistent with reduced informed trading. Lastly, we document a positive effect of GenAI-assisted trading on long-run stock price informativeness. Overall, our findings indicate that a significant number of investors use ChatGPT in ways that influence their trading decisions and market outcomes. Future research can investigate the mechanisms underlying these GenAI effects and the potential risks of using GenAI for trading.

1. Introduction

The rise of generative artificial intelligence (GenAI) has transformed how individuals process information. With advanced natural language capabilities, GenAI models serve as general-purpose tools for a wide range of tasks—such as information searches, text analysis, and content creation—while enabling users to customize outputs via context-based conversations. As a prime example of GenAI models, ChatGPT was launched in November 2022 and has since made advanced GenAI tools broadly accessible. The combination of versatility, adaptability, and broad accessibility distinguishes GenAI models from traditional computer-based tools and has generated widespread enthusiasm about their potential to boost efficiency and productivity. Nascent research demonstrates ChatGPT's potential to facilitate investment-related tasks, such as analyzing corporate disclosures and forecasting earnings (e.g., Bertomeu et al., 2025; Bradshaw et al., 2025; Li et al., 2023). Yet, there is little research on the real-world impact of ChatGPT on investors' trading activities.

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In this paper, we use ChatGPT outages to provide early evidence on two related questions: (1) To what extent do investors rely on GenAI for professional tasks? (2) How does GenAI-assisted trading affect stock price informativeness? The motivation for this investigation is two-fold. First, our investigation can shed light on the prevalence of GenAI uptake in the financial sector after the release of ChatGPT. Investors may favor ChatGPT over existing tools due to its ability to handle a wide range of tasks within a single interface through natural language instructions. In addition, ChatGPT's broad accessibility implies that a wide range of investors, including retail investors, can use it for professional tasks. However, concerns over information accuracy and data security might discourage its adoption, as evidenced by several investment banks, including Bank of America and Goldman Sachs, reportedly restricting their employees' access to ChatGPT in early 2023.¹ At the time of writing, many firms were deliberating the potential effects of ChatGPT on their operations and the extent of its adoption.² As such, an empirical evaluation of its real-world applications can yield valuable insights. Second, our investigation can also provide insights for regulators who are concerned about the risks that AI-based trading models pose to financial markets. The Financial Stability Oversight Council identified AI as "an emerging vulnerability in the financial system" in December 2023 and highlighted the risks associated with erroneous or inconsistent outputs from GenAI models.³ Investors' dependence on the processing capabilities of GenAI models could lead to market instability if these models do not provide accurate or verifiable information and analysis.

Our review of anecdotal evidence suggests that investors use ChatGPT for a diverse set of information acquisition, information integration, and routine tasks (see Appendix A for details). Regarding information acquisition, investors can use ChatGPT to obtain information on and analyses of specific stocks, industries, and broad market conditions. They can also use ChatGPT to quickly extract and summarize relevant information from external sources. For information integration, ChatGPT can assist investors in analyzing news sentiment, forecasting future performance, recommending trading directions, and evaluating portfolio allocations. Lastly, ChatGPT can support investors with routine tasks, such as email drafting, report preparation, document review, and coding, which can increase the time available for trading. Because investors' use of ChatGPT is not observable, an empirical analysis is necessary to assess the overall extent to which investors rely on the tool for professional tasks and its resulting impact on financial markets.

We rely on ChatGPT outages as plausibly exogenous shocks to the availability of GenAI tools to investors and examine how trading activities change during these periods. We collect information on ChatGPT outages from OpenAI's official incident reports, supplemented with user reports from DownDetector. In total, we identify eight major outage events that occurred during trading hours between February and August 2023. These outages range from 18 min to 4 h in duration, with an average of 119 min. The outages took place on different dates and at different times of the day, reducing the likelihood that any observed changes in trading activities are driven by date- or time-specific market conditions. We also search related news to confirm that the outages were not coincident with other service outages.

We first examine changes in stock trading volume during ChatGPT outage periods. If investors rely on ChatGPT for professional tasks, an unexpected outage of ChatGPT would disrupt their workflow and reduce their capacity to carry out trading-related activities.⁴ As suggested by prior research (e.g., Hirshleifer et al., 2009; Brown et al., 2015; deHaan et al., 2017), investors with limited capacities often delay or forego trading decisions, resulting in reduced stock market activities. This effect is likely to be more pronounced when investors need to make trading decisions within a short time frame. However, if investors do not rely on ChatGPT for professional tasks, or if they can seamlessly substitute it with alternative tools, we would not observe any change in trading volume during ChatGPT outage periods.

To assess changes in trading activities, we compare trading volume during outage periods with that during the same periods over the previous five trading days (hereafter, control periods) for the same firm. Using intraday data from the New York Stock Exchange's Trade and Quote (TAQ) database, we calculate trading volume for each 5-min interval and document a significant decline in trading volume during the outage periods relative to the control periods after controlling for firm characteristics, firm fixed effects, and 5-min interval fixed effects. This finding indicates that investors utilize ChatGPT in ways that influence their trading decisions. The effect is also economically important: the drop in trading volume is approximately 5.55 % of the sample standard deviation.⁵

These results are robust to using alternative measures of trading volume, controlling for more stringent fixed effects, employing alternative control periods, controlling for the effect of extreme values, and using alternative start times of outages. We also conduct a falsification test using the day before the outage event as the pseudo-outage date and do not find a similar effect. Importantly, our analysis of dynamic changes in trading volume surrounding outages shows an insignificant effect in the 1-h window immediately preceding or following outages. In addition, we observe a significant decrease in trading volume beginning about 30 min after the start of outages. This result suggests that ChatGPT is likely to be used for information processing and other routine tasks rather than for

¹ See <https://www.forbes.com/sites/brianbushard/2023/02/24/workers-chatgpt-use-restricted-at-more-banks-including-goldman-citigroup>.

² For example, see <https://www.wsj.com/articles/generative-ai-promises-an-economic-revolution-managing-the-disruption-will-be-crucial-b1c0f054>, <https://www.bloomberg.com/news/articles/2023-03-20/using-chatgpt-at-work-nearly-half-of-firms-are-drafting-policies-on-its-use>.

³ See <https://home.treasury.gov/system/files/261/FSOC2023AnnualReport.pdf>.

⁴ Even if investors primarily use ChatGPT for routine tasks that are not directly related to trading, an outage could increase the time required to complete those tasks, thereby reducing the time available for trading activities. This argument is consistent with the "task complementarity" effect discussed in Acemoglu (2024).

⁵ In the main analyses, we focus on ChatGPT, the interactive application designed for human users because we are more interested in the interactions between human users and GenAI tools. In an additional analysis, we examine the impact of the outages of OpenAI's application programming interface (API) on trading activities. We find a significant drop in algorithmic trading volume during API outages, suggesting that OpenAI's API services are likely integrated into certain trading algorithms and thus can influence algorithmic trading.

direct trading activities; otherwise, we would observe an immediate decline in trading volume at the start of outages.

We conduct two additional analyses to explore the potential heterogeneous effect of ChatGPT on investors' trading activities. First, we observe a larger decline in trading volume during outages for firms with corporate news released on the same day as the outage but before its end time. This result suggests that the effect of ChatGPT outages is more pronounced when investors have a greater need for information processing. Second, we disaggregate trading volume into retail and non-retail trading volume and find significant declines in both retail and non-retail trading volume during ChatGPT outages, with a more pronounced effect for non-retail trading volume. We further document a larger decline in trading volume for firms with higher ownership by transient institutional investors. These findings suggest that ChatGPT outages have a more pronounced impact on investors who rely on it for timely information processing and trading decisions.

We turn to our second research question to understand the impact of GenAI tool adoption on market quality and price informativeness. We start by examining changes in market quality during ChatGPT outages. First, we observe a significant drop in short-term stock price impact and return variance, consistent with a reduction in informed trading during ChatGPT outages, suggesting that ChatGPT usage facilitates informed trading. Second, we find a decrease in bid-ask spreads, consistent with a reduction in information asymmetry among investors during ChatGPT outages. This result, taken together with the previous findings of non-retail investors' greater reliance on ChatGPT than retail investors, suggests that ChatGPT may not level the playing field but instead enhances the information advantage of institutional investors over retail investors. However, the effects observed in both tests are relatively modest in economic magnitude, suggesting that ChatGPT adoption was either not yet widespread or offered limited advantages during our sample period.

While our short-run analyses using ChatGPT outages offer identification advantages, they do not necessarily indicate long-run benefits of using ChatGPT for trading activities. If investors use ChatGPT merely as a more convenient replacement for pre-existing tools, the observed short-run effects during outages may simply reflect switching costs—difficulties in quickly switching to alternative tools—and do not imply that ChatGPT offers additional benefits over those tools in the long run. To provide early evidence on this issue, we conduct a difference-in-differences analysis surrounding the launch of ChatGPT. For each firm, we use the magnitude of its trading volume decline during ChatGPT outages to proxy for its GenAI-assisted trading intensity. We use the price informativeness measure developed by [Bai et al. \(2016\)](#) and [Kacperczyk et al. \(2021\)](#) to capture the extent to which a firm's current stock price predicts its future earnings. We document that firms with higher GenAI-assisted trading intensity experience a greater improvement in price informativeness following the launch of ChatGPT. While the selection issue associated with our proxy for GenAI-assisted trading intensity prevents us from drawing causal inferences, these tests provide suggestive evidence consistent with ChatGPT facilitating investors' informed trading in the long run.

This paper contributes to the literature by providing robust evidence of economically meaningful declines in trading volume during ChatGPT outages and early evidence on the improvement in long-run stock price informativeness after the launch of ChatGPT. Our results indicate that a significant number of investors use ChatGPT in ways that influence their trading decisions shortly after its launch. As ChatGPT has been criticized for its tendency to generate false information, our findings suggest that the benefits of using ChatGPT in trading may outweigh its potential risks. The evidence in this paper can thus help both financial institutions and regulators assess the effects of GenAI models on financial markets.

Our paper adds to the emerging literature on GenAI. Most concurrent studies focus on the applications of ChatGPT in processing and generating insights from text information,⁶ whereas few studies explore the real-world impact of ChatGPT usage. Our study is closely related to that of [Bertomeu et al. \(2025\)](#), who examine the impact of the ban on ChatGPT in Italy on analysts' forecasting activities. In addition, [Blankespoor et al. \(2024b\)](#) examine firms' use of GenAI in drafting financial disclosures. Unlike these studies, we examine the disruptive effects of ChatGPT outages on investors' trading activities. This analysis is particularly relevant given ChatGPT's role as a public, general-purpose tool, which may amplify the impact of its outages on trading compared to earlier periods when investors relied on more specialized, task-specific tools.

Our paper also adds to the literature on the use of new technologies in capital markets. [Blankespoor, deHaan, and Marinovic \(2020\)](#) emphasize the fundamental role of technological advancement in reshaping investors' information processing activities and argue that the effects might vary across investor groups and technology types.⁷ They call for more research to examine how the increasingly sophisticated technologies affect different groups of investors and market outcomes. While prior studies have documented AI's advanced capabilities in certain investment-related tasks (e.g., [Cao et al., 2024](#)), to the best of our knowledge, our study is among the first to examine whether investors use GenAI tools in ways that influence their trading activities and how such usage varies across investor groups. Although GenAI's broad accessibility could be seen as benefiting retail investors more, our evidence suggests that its benefits appear to be greater for institutional investors—particularly transient institutional investors—than for retail investors.

Our study motivates several questions for future research. First, due to data limitations, we are unable to observe *how* investors use ChatGPT. Future research can use survey or field data to shed light on this issue and help us understand whether investors rely on GenAI tools for non-trading routine tasks, information processing tasks, and/or direct trading decisions, such as generating trading recommendations. Future research can also examine which of these non-exclusive mechanisms plays a more important role in

⁶ For examples, see [de Kok \(2025\)](#), [Hansen and Kazinnik \(2023\)](#), [Li et al. \(2023b\)](#), [Lopez-Lira and Tang \(2023\)](#), [Li et al. \(2023a\)](#), and [Jha et al. \(2024\)](#).

⁷ For example, while broader dissemination of corporate disclosures via the SEC's EDGAR helps individual investors make more informative trades ([Gao and Huang, 2020](#)), the adoption of XBRL benefits institutional investors more than individual investors, increasing the information asymmetry among investors ([Blankespoor et al., 2014](#)).

explaining our results. Second, while we provide suggestive evidence that ChatGPT benefits institutional investors more than retail investors, future research can employ more direct measures of retail and institutional trading to examine the level of GenAI adoption across investor types and to better understand its impact on the information advantage of sophisticated investors over others. Third, our analysis of long-run stock price informativeness relies on an ex-post measure to capture a firm's GenAI-assisted trading intensity, which is inherently noisy and potentially subject to selection bias. Thus, a field or lab experiment that randomly grants investors access to GenAI tools could help better identify causal effects. Finally, given the rapid evolution of GenAI models and their expanding use cases among investors, their effects on market outcomes could change over time, which can be examined using more recent GenAI outages. It remains an open question whether the disruptive effect of ChatGPT outages intensifies with its wider adoption in more recent periods, or weakens as alternative GenAI tools become more widely available. Nevertheless, the results can inform regulators who are concerned that investors' overreliance on one GenAI tool pose an emerging vulnerability in the financial system.

2. Institutional background and related research

2.1. Institutional background

OpenAI launched ChatGPT as an interactive chatbot on November 30, 2022. ChatGPT was built on a large language model and can provide answers to users in a conversational style. Following the introduction of ChatGPT, several technology giants released similar products. As of August 2023, ChatGPT led the market with a total of 14.6 billion website visits, nearly 60 times the number of visits received by Alphabet's Bard (now known as Gemini).⁸

ChatGPT's rapid rise in popularity is largely driven by its versatility, adaptability, and broad accessibility, which distinguish it from traditional computer-based tools. First, ChatGPT is a general-purpose tool that can perform a wide range of tasks without requiring specialized set-up or prior knowledge. This versatility greatly improves its capability to tackle real-world investing problems, which often involve multiple types of tasks. Second, ChatGPT provides users with considerable flexibility to interact with the tool via conversational languages, allowing users to tailor output to meet specific needs. Third, prior to the launch of ChatGPT, certain advanced tools were only available to select investors due to their costs and proprietary nature. The launch of ChatGPT has made such capabilities accessible to a broader set of investors. Collectively, these three attributes of ChatGPT can lead to investors' reliance on the tool for a wide range of professional tasks.

Some finance professionals are reported to have used ChatGPT in their work since its debut. Survey evidence from Ebert (2023) indicates that about half of the 39 surveyed fund managers have utilized ChatGPT for professional tasks, such as analyzing regulatory filings, summarizing research reports, and drafting documents and emails. Several reports from the business press also confirm the ongoing or planned adoption of ChatGPT within prominent asset management firms, such as Citadel and Bridgewater Associates.⁹ Recent research also ranks finance jobs among the top occupations that are exposed to ChatGPT (Eisfeldt et al., 2023). Appendix A presents a list of anecdotal evidence that illustrates potential use cases of ChatGPT by investors. As illustrated by the list, ChatGPT can support a wide range of information-processing tasks that are relevant to trading decisions. However, there are only a few cases of its direct use in generating trading recommendations, likely due to its potential limitations discussed below.

Employing ChatGPT for financial tasks has limitations and is controversial, and not all investors are receptive to it, particularly in the period shortly after its release. According to a Bloomberg interview in February 2023, some traders are skeptical about the accuracy and timeliness of its output and its ability to formulate sensible trading recommendations.¹⁰ In addition, following the release of ChatGPT, a few investment banks have restricted employees' use of ChatGPT due to concerns about inaccurate information and the potential leakage of sensitive information.¹¹ Consistent with these concerns, research finds that ChatGPT is subject to "hallucination," where the tool "invents" incorrect information and presents it as factual (Athaluri et al., 2023; Emsley, 2023). In addition, ChatGPT faces challenges in handling mathematical operations, which are essential to financial analysis.¹²

The above discussions imply that it is an empirical question whether a significant number of investors adopt ChatGPT for professional tasks in ways that influence their trading decisions and market outcomes.

⁸ See <https://writerbuddy.ai/blog/ai-industry-analysis>.

⁹ For example, in March 2023, the founder and CEO of Citadel, Kenneth Griffin commented that "(t)his branch of technology has real impact on our business. Everything from helping our developers write better code to translating software between languages to analyz(ing) various types of information." In July 2023, Greg Jensen, the Co-Chief Investment Officer at Bridgewater Associates, commented on ChatGPT's capability in handling questions that Bridgewater asked its investment associates and revealed that Bridgewater was integrating large language models with statistical models to inform their trading decisions (<https://www.bloomberg.com/news/articles/2023-07-03/bridgewater-s-greg-jensen-explains-how-the-world-s-biggest-hedge-fund-is-investing-in-ai>).

¹⁰ See <https://www.bloomberg.com/news/articles/2023-02-24/wall-street-s-chatgpt-nightmare-scenario-gets-a-reality-check>.

¹¹ See <https://www.forbes.com/sites/brianbushard/2023/02/24/workers-chatgpt-use-restricted-at-more-banks-including-goldman-citigroup/?sh=668f10716cf4>.

¹² See <https://www.wsj.com/articles/chatgpt-openai-math-artificial-intelligence-8aba83f0>.

2.2. Related research

Emerging research in accounting and finance focuses on ChatGPT's usefulness in processing natural language in various settings.¹³ For example, some recent studies use ChatGPT to create business summaries from corporate disclosures (Breitung and Müller, 2025), predict future earnings using earnings press releases (Li et al., 2023b), and analyze corporate culture based on conference call transcripts and analyst reports (Li et al., 2023b). Several studies use ChatGPT to process business news (Lopez-Lira and Tang, 2023), social media posts (Cho, 2024), and government statements (Hansen and Kazinnik, 2023). Collectively, this body of research demonstrates that ChatGPT has the capability to support a wide range of investment-related tasks.

Our study complements prior research by investigating the early use of ChatGPT by investors and the associated impact on stock price informativeness. Trading decisions require investors to acquire and integrate relevant information, estimate a security's intrinsic value, and forecast its future returns. It is well documented that corporate information releases represent key opportunities for skilled investors to quickly process information and make profitable trades (e.g., Huang et al., 2020; Fedyk, 2024). Institutional investors, in particular, have been shown to excel at processing financial information (e.g., Ke and Ramalingegowda, 2005). The evidence underscores the importance of timely information processing, especially for institutional investors, in making effective trading decisions, and supports the potential role of ChatGPT in facilitating investors' trading decisions.¹⁴ We take advantage of ChatGPT's unexpected outages during stock trading hours to investigate the effect of ChatGPT usage on investors' trading activities. If investors rely on ChatGPT for professional tasks, an unexpected outage of ChatGPT would disrupt their workflow and reduce their capacity to carry out trading-related tasks, resulting in reduced trading activities. We further rely on the decrease in trading volume during ChatGPT outages to construct a firm-level measure of GenAI-assisted trading intensity and examine its impact on stock price informativeness in the long run.

3. Data and sample

We identify a set of ChatGPT outage events between February and August 2023 based on official reports from OpenAI. OpenAI maintains a public record of the date and severity of each ChatGPT outage since February 14, 2023, along with detailed incident reports outlining the timeline and nature of each incident.¹⁵ We begin with all incidents that involve major outages, during which ChatGPT was not accessible to users.¹⁶ For each incident, we collect the time when OpenAI began to investigate the malfunction and the time when it was fully resolved. In total, we identify eight major ChatGPT outages that coincided at least partially with U.S. stock trading hours. To better identify each outage's start time, we rely on the user reports compiled by DownDetector of Ookla, a widely used platform that tracks online service disruptions. The data has been used in prior research to identify brokerage platform outages (e.g., Eaton et al., 2022). We aggregate the number of user reports about OpenAI service issues at 5-min intervals and define the start time of each outage as the first interval in which the number of reports exceeds the sample mean by more than one standard deviation based on the sample for the same 5-min interval over the period of February to August 2023.¹⁷ We use the time when the outage was fully resolved by OpenAI as the end time. Appendix B provides the list of the eight outages and the relevant details. The outages happened on eight different days, with durations ranging from 18 to 240 min. These outages occurred at various times of day and spanned all days of week.

Although OpenAI does not disclose the specific causes of each outage, we are able to identify the causes of three outages based on OpenAI's incident reports and our own search for related news. These causes vary and include instability in the underlying database, misconfigurations in program settings, and bugs in the software. We also conduct a comprehensive search of related news and do not find any systematic evidence of other major service outages that occurred during the same periods.¹⁸ The idiosyncratic causes suggest that ChatGPT outages are unlikely to coincide with economy-wide disruptions that would also affect investors' trading activities.

We obtain intraday trading data from the NYSE's TAQ database. We collect daily stock prices from the Center for Research in Securities Prices (CRSP), companies' accounting information from Compustat, news coverage and press release data from RavenPack, SEC filing data from the Wharton Research Data Services SEC Analytics Suite, institutional holdings from the Thomson 13F filings, and management and analyst forecasts from the Institutional Brokers' Estimate System (I/B/E/S).

We start with all securities with available TAQ data. For each security, we aggregate its intraday transactions to 5-min intervals as

¹³ See de Kok (2025) for a discussion of the applications of ChatGPT in textual analysis and a comparison of ChatGPT with other methods.

¹⁴ Our review of books written by industry practitioners also supports the view that information processing efficiency is crucial in the fast-paced trading environment. For example, Bellaïeff (2010), co-founder of SMB Capital, comments that "In our business, 5 s can be eternity, so the ability to process information quickly and react appropriately is crucial." He also emphasizes the importance of trading on corporate news, stating that "We make sure our traders focus on stocks with fresh news (Stocks In Play)," and "Stocks In Play offer us more opportunities, and trades that we have to hold for only short periods of time."

¹⁵ See <https://status.openai.com/uptime> for ChatGPT's uptime records and <https://status.openai.com/history> for OpenAI's incident history.

¹⁶ Some of these incidents also include certain periods of partial outages, which we are unable to isolate due to data limitations.

¹⁷ The mean and standard deviation are calculated based on our analysis of DOWNDetector® data from February to August 2023. Ookla® trademarks are used under license and reprinted with permission.

¹⁸ We systematically investigate three types of potential outages: (1) outages of alternative large language models, including Claude, Gemini (formerly Bard), Character.AI, Llama, and Mistral AI; (2) outages of cloud services, including Amazon Web Services, Microsoft Azure, and Google Cloud; and (3) outages of other key technology and internet services, including the SEC's EDGAR platform, Internet providers (e.g., T-Mobile), and Twitter.

in prior research (e.g., [Eaton et al., 2022](#)). For each ChatGPT outage, we include all of the 5-min interval observations in the corresponding trading periods, as well as all of the 5-min interval observations during the same time periods of the previous five trading days.¹⁹ Our sample selection starts with the maximum number of 11,645,568 possible observations, which is calculated as 192 5-min intervals across the eight outage events, multiplied by six days and 10,109 unique securities for each outage. We exclude observations on the control days that coincide with other ChatGPT outages. We then exclude securities without CRSP data on stock prices, securities that are not ordinary common stocks, and securities that are not traded on the NYSE, AMEX, or Nasdaq. We further drop observations of securities with no trading activity on the day, which occurs because of illiquidity, potential data errors in the TAQ database, or trading suspensions.²⁰ We then drop observations with closing stock prices below US\$5 on the day preceding the outages. Lastly, we drop observations with missing control variables. Our final sample consists of 2,615,434 security-date-5-minute interval observations from 2553 unique stock securities.²¹ [Table 1](#) summarizes the sample selection process.

4. Changes in trading volume during ChatGPT outages

4.1. Research design

We use the following ordinary least squares regression model to examine the impact of ChatGPT outages (*Outage*) on trading volume (*Trading Volume*):

$$\text{Trading Volume}_{i,t,p} = \beta_0 + \beta_1 \text{Outage}_{t,p} + \gamma \text{Controls}_{i,t} + \theta_i + \eta_p + \varepsilon_{i,t,p}, \quad (1)$$

where i , t , and p indicate firm, date, and 5-min interval, respectively. *Trading Volume* _{i,t,p} is defined as the total number of shares traded in the 5-min interval p of day t , divided by the total number of shares outstanding for firm i on day t , then multiplied by 10,000 for ease of presentation. The independent variable of interest, *Outage* _{t,p} , equals one if time interval p of day t falls in a ChatGPT outage period, and zero otherwise. The coefficient on *Outage* _{t,p} , β_1 , captures the difference in trading volume between ChatGPT outage periods and the same periods of the preceding five trading days. If a ChatGPT outage disrupts investors' trading decisions, we should observe a decrease in trading volume during the outage period and thus a negative coefficient on *Outage*.

We include a set of firm characteristics that may affect trading volume, as shown in prior research (e.g., [Eaton et al., 2022](#)): overnight stock returns (*Return Overnight*), returns in the previous day (*Return [-1]*), cumulative returns from day $t-5$ through day $t-2$ (*Return [-5, -2]*), price range in the previous 60 days (*Price Range*), stock prices on day $t-1$ (*Price*), and market capitalization on day $t-1$ (*Market Cap*). We winsorize all continuous variables, except stock returns and prices, at the 1 % tails.²² We also include firm fixed effects (θ_i) to control for time-invariant firm characteristics and 5-min interval fixed effects (η_p) to control for time-varying market-wide conditions across different times in a day. The t -statistics are based on standard errors clustered at the firm and date levels. [Appendix C](#) provides detailed definitions of the variables.

[Table 2](#) reports the summary statistics of the variables. The mean of *Trading Volume* in a 5-min interval is 0.850, i.e., 0.00850 % of total shares outstanding. The mean of *Outage* is 0.170, indicating that 17.0 % of the observations are from outage periods. The descriptive statistics of the control variables are largely similar to those reported in prior research (e.g., [Crawford et al., 2012](#); [Eaton et al., 2022](#)).

4.2. Main results and sensitivity tests

4.2.1. Main results

[Table 3](#) reports the baseline results. Column (1) includes firm and 5-min interval (i.e., Time Interval) fixed effects. The coefficient on *Outage* is significantly negative ($t = -2.62$). Column (2) further includes control variables. The coefficient on *Outage* remains significantly negative ($t = -3.25$). These results suggest a decrease in trading volume during the ChatGPT outage periods relative to the control periods. This effect is economically significant; the coefficient (-0.084) in Column (2) suggests a relative decrease of 5.55 % ($= -0.084/0.850$) of the sample standard deviation of *Trading Volume*.²³

¹⁹ For example, for the first ChatGPT outage, which occurred on February 14, 2023 and lasted from 13:55 to 14:37, we include nine 5-min interval treatment observations from February 14 and nine observations for the same 5-min intervals from each of the preceding five trading days (February 7, 8, 9, 10, and 13).

²⁰ We identify the potential reasons for the lack of trading in the TAQ database by cross-checking this set of observations against public databases of trading halts and daily stock data from the CRSP. The results are quantitatively similar when we include these observations and set their trading volume to be zero (untabulated).

²¹ The maximum possible number of observations for the 2553 securities is 2,941,056 ($= 2553 \times 192 \times 6$). The reduction in sample size is due to the removal of the following observations: (1) 53,613 observations on the control days coinciding with other ChatGPT outages, (2) 31,584 observations with missing stock price on the date preceding the outage, (3) 20,681 observations without trading activity, (4) 192,369 observations with a pre-outage price below US\$5, and (5) 27,375 observations with missing control variables.

²² Our results remain similar if we also winsorize stock returns and prices at the 1 % tails.

²³ The decrease is about 9.88 % ($= -0.084/0.850$) of the sample mean of *Trading Volume*. In comparison, [Brown et al. \(2015\)](#) documents a 1.87 % decrease in trading volume when mobile communication is restricted during commutes following the enactment of distracted driving laws (see their [Table 6](#), Column 2), suggesting that ChatGPT outages have a larger effect.

Table 1
Sample Selection

This table presents the sample selection process. The final sample for the trading volume analyses includes 2,615,434 security-date-5-minute interval observations from 2553 stock securities during the period from February to August 2023.

	# Observations	# Securities
Initial sample: Intraday 5-min intervals during the ChatGPT outage periods and the same time intervals for the same firm in the preceding five trading days (control periods)	11,645,568	10,109
Less:		
Intervals in the control periods that coincide with other outages	(212,289)	(0)
Observations with missing CRSP data on stock prices	(4,966,556)	(4298)
Non-ordinary common stocks and stocks that are not listed on NYSE, AMEX, or Nasdaq	(2,627,194)	(2364)
Observations with no trading activity on the day	(28,507)	(0)
Observations with closing stock prices on the day preceding the outage lower than US\$5	(1,168,213)	(894)
Observations with missing control variables	(27,375)	(0)
Final sample	2,615,434	2553

Table 2
Descriptive Statistics of Variables Used in Trading Volume and Trading Impact Analyses

This table reports the descriptive statistics of the variables used in the trading volume and trading impact analyses. The sample includes 2,615,434 security-date-5-minute interval observations from 2553 stock securities during the period from February to August 2023.

Variable	N	Mean	SD	p25	p50	p75
Variables used in the main trading volume analyses						
<i>Trading Volume</i>	2,615,434	0.850	1.513	0.117	0.368	0.874
<i>Outage</i>	2,615,434	0.170	0.375	0.000	0.000	0.000
<i>Return Overnight</i>	2,615,434	−0.001	0.030	−0.007	0.000	0.005
<i>Return [−1]</i>	2,615,434	−0.003	0.037	−0.016	−0.001	0.010
<i>Return [−5, −2]</i>	2,615,434	−0.010	0.057	−0.033	−0.008	0.012
<i>Price Range</i>	2,615,434	0.218	0.142	0.128	0.186	0.266
<i>Price</i>	2,615,434	3.575	1.085	2.699	3.476	4.323
<i>Market Cap</i>	2,615,434	14.431	1.945	13.032	14.385	15.666
Dependent variables used in additional analyses						
<i>Shares Count</i>	2,615,434	12,668	33,017	374	2494	9132
<i>Trades Count</i>	2,615,434	171	296	15	70	188
<i>Retail Volume</i>	2,615,434	0.057	0.158	0.000	0.008	0.039
<i>Non-Retail Volume</i>	2,615,434	0.787	1.393	0.106	0.343	0.816
<i>Odd-Lot Volume</i>	1,456,687	0.187	0.276	0.031	0.101	0.222
<i>Price Impact</i>	2,408,754	0.072	0.203	−0.002	0.025	0.098
<i>Return Variance</i>	2,169,531	0.074	0.127	0.021	0.036	0.069
<i>Bid-Ask Spread</i>	2,386,031	0.426	0.776	0.081	0.164	0.374

4.2.2. Sensitivity tests

We conduct numerous sensitivity tests to ensure the robustness of our baseline results. First, we adopt two alternative measures of trading volume: the total number of shares traded (*Shares Count*) and the total number of trades (*Trades Count*) in each 5-min interval. As these variables are count-based, we follow the suggestion of [Correia et al. \(2019\)](#) and [Cohn et al. \(2022\)](#) to use a Poisson pseudo maximum likelihood regression model with high-dimensional fixed effects. Panel A of [Table 4](#) reports the regression results. The coefficient on *Outage* is significantly negative at the 1 % level in both cases.

Second, we employ three sets of more stringent fixed effects: (1) outage $\times$ firm and outage $\times$ 5-min interval fixed effects to control for outage-specific effects of firm and time-interval factors²⁴; (2) outage $\times$ firm $\times$ 5-min interval fixed effects to isolate the variation in trading volume across different days for the same outage event, the same firm, and the same time interval; (3) day of week and firm joint fixed effects and day of week and 5-min interval joint fixed effects, further controlling for the variation by day of week. As reported in Panel B of [Table 4](#), the coefficient on *Outage* is negative and significant at the 5 % level in all cases.

Third, to ensure that our results are robust to alternative control days, we use the 10 or 15 trading days (instead of 5 days) preceding ChatGPT outages as the control periods. As reported in Panel C of [Table 4](#), the inferences remain the same. In addition, to ensure that our results are not affected by the possibility that outages are caused by investors' excessive usage of ChatGPT on the days immediately before outages, we use trading days [−10, −6] or [−15, −6] as the control periods. Again, as reported in Panel C of [Table 4](#), the

²⁴ Even though the result appears to be stronger in terms of the magnitude of the coefficient based on this specification, we do not use it as our main specification due to concerns that high-dimensional fixed effects may inflate coefficient estimates as suggested by [Armstrong et al. \(2022\)](#) and [Jennings et al. \(2024\)](#). Nonetheless, we confirm that our results in [Tables 5–8](#) remain qualitatively similar when using this specification (Untabulated).

inferences remain the same.

Fourth, to ensure that our inferences are not driven by outliers, we estimate Equation (1) using a median regression. We also estimate Equation (1) after excluding observations with residuals in the top/bottom 1 % or 2.5 %. As reported in Panel D of Table 4, while the results based on these tests are weaker, the inferences remain the same.

Fifth, to the extent that shorter ChatGPT outages likely have a smaller impact on investors' trading activities, we drop the shortest, the two shortest, or the three shortest ChatGPT outages from the analysis. As reported in Panel D of Table 4, the results are quantitatively similar.²⁵

Sixth, to ensure that our results are not sensitive to the identification of outages' start times based on user reports, we use OpenAI's start time of investigation as each outage's start time. As reported in Panel E of Table 4, the results remain similar.

Lastly, we conduct a falsification test using day $t-1$ as the pseudo-outage day and days $t-2$ through $t-6$ as the corresponding control days. Panel E of Table 4 presents the regression results from this falsification test. We do not find a significant drop in trading volume during the pseudo-outage periods relative to the corresponding control periods, alleviating the concern that our main findings are driven by certain market-wide factors surrounding the outage days.

4.2.3. Dynamic analysis

To shed light on when ChatGPT outages start to affect investors' trading activities, we conduct a dynamic analysis. Following Eaton et al. (2022), we focus on outages that begin after 10:00 a.m. and end before 3:30 p.m. to ensure sufficiently long pre- and post-periods on the same trading day. We separately estimate the effects for every 10-min window surrounding ChatGPT outages and plot the coefficient estimates along with their 90 % confidence intervals in Fig. 1. The negative effect on trading volume starts to become significant after 30 min, suggesting that it takes time for ChatGPT outages to affect investors' trading decisions.²⁶ The results support the notion that ChatGPT is more likely to be used for information processing rather than for later stages of the trading process (e.g., direct trading execution); otherwise, we would observe an immediate decline in trading volume after the outages start.²⁷

In addition, there does not appear to be a "catching up" in trading volume after the outages end. Ex ante, we do not expect to find strong evidence on "catching up" because a ChatGPT outage only affects a subset of market participants who rely on the tool for professional tasks. Our finding of a 5.55 % decline in trading volume during outages is consistent with this point. By the time the outage ends, affected investors may find that stock prices have already incorporated the relevant information, reducing the necessity of trading. However, our empirical setting does not allow us to provide direct evidence for such a claim.

4.3. Mechanism analyses: corporate news release

We conduct cross-sectional analyses to shed light on the potential mechanisms through which ChatGPT assists investors with their trading activities. The survey and anecdotal evidence presented in Appendix A suggests that investors mainly use ChatGPT for information processing activities. If so, we would expect that the effect of ChatGPT outages on trading volume is stronger when investors have a greater need for information processing.

Following Drake, Roulstone, and Thornock (2012), we capture investors' demand for assistance with information processing activities using the presence of corporate news (*News*). For observations on outage days, *News* is an indicator variable that equals one if at least one corporate announcement (i.e., earnings announcement, management forecast, press release, or SEC filing), news article from RavenPack, or analyst forecast about the firm is released on the outage day but before the end of the outage period, and zero otherwise. *News* is similarly defined for observations on control days, based on whether any corporate news is released on the control day prior to the end of the corresponding outage period. We then examine the incremental effect of ChatGPT outages on trading volume when there is news released on the day of the outage but before the end of the outage period.²⁸

Table 5 reports the regression results. Column (1) includes control variables and firm and 5-min interval fixed effects. Column (2) adopts a fully saturated model by further including the interaction terms of *News* with all of the control variables and fixed effects,

²⁵ Six of the eight outage events in our sample occurred after the release of ChatGPT-4 on March 13, 2023. Untabulated analysis suggests that our inferences hold both before and after the release of ChatGPT-4, and we do not find a significant difference in the results between these periods.

²⁶ We also conduct the dynamic analysis using all outages. For outages where the 1-h pre- or post-outage window is not within the trading hours of the same day, we extend the window to the preceding or subsequent trading day to estimate the corresponding coefficients. These windows may not be comparable because they occur either near the closing of the previous trading day or at the opening of the following trading day. Nonetheless, we observe similar patterns, with the coefficient becoming significant 30 min after the start of outages.

²⁷ In an untabulated analysis, we document a significant increase in the access of SEC filings in the HTML format (a proxy for human access) via the EDGAR platform during ChatGPT outages, particularly for the four major filing types (i.e., 10-K, 10-Q, 8-K, and DEF 14 filings) that contain detailed information about firms, for older filings that are released before the outage or control day, and for longer outages. These results are consistent with two non-exclusive explanations: (1) investors rely on EDGAR as an alternative information source to obtain firms' general and/or historical information during ChatGPT outages; (2) investors retrieve historical filings from EDGAR needed for the analysis when ChatGPT resumes its services. Both arguments suggest that investors likely experience a decrease in the efficiency of information processing during ChatGPT outages, resulting in a decrease in their trading activities.

²⁸ Our inferences remain the same if we only include news released *during* the outage period. Separately, in an untabulated analysis, we create two indicator variables for the news that is released on the first day and second day prior to the outage date, respectively, and add their interaction terms with *Outage* to the regression. We find insignificant coefficients on both interaction terms, suggesting that the effect of ChatGPT outages on trading volume is more pronounced for firms with news released on the outage days but not for firms with news released on the earlier days.

Table 3
ChatGPT Outages and Change in Trading Volume

This table reports the regression results of the analysis of the impact of ChatGPT outages on trading volume at 5-min intervals using the following model:

$Trading\ Volume_{i,t,p} = \beta_0 + \beta_1 Outage_{i,t,p} + \gamma Controls_{i,t} + \theta_i + \eta_p + \varepsilon_{i,t,p}$ The sample includes 2,615,434 security-date-5-minute interval observations from 2553 stock securities during the period from February to August 2023. See Appendix C for variable definitions. Standard errors are clustered at the firm and day levels. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively, based on two-sided tests.

Dependent Variable =	Trading Volume	
	(1)	(2)
<i>Outage</i>	−0.088** (−2.62)	−0.084*** (−3.25)
<i>Return Overnight</i>		0.536 (0.51)
<i>Return [−1]</i>		−0.248 (−0.44)
<i>Return [−5, −2]</i>		−0.682 (−1.63)
<i>Price Range</i>		1.489*** (8.52)
<i>Price</i>		0.276** (2.18)
<i>Market Cap</i>		−0.285** (−2.68)
Firm FE	YES	YES
Time Interval FE	YES	YES
N	2,615,434	2,615,434
Adjusted R ²	0.446	0.454

following the suggestion of deHaan et al. (2023).²⁹ The coefficient on *Outage* continues to be significantly negative in both columns.³⁰ More importantly, consistent with our expectation, the coefficient on *Outage* × *News* is significantly negative in both columns ($t = -1.72$ and -2.54 , respectively). These results indicate a larger decline in trading volume for firms with identified firm-specific news released right before or during ChatGPT outage periods, consistent with the notion that ChatGPT can help investors process firm-specific information.

4.4. Trading by different types of investors

4.4.1. Retail versus institutional investors

Next, we analyze the change in trading volume from retail and institutional trades separately to understand which investor group utilizes ChatGPT more. Ex ante, it is unclear which investor group benefits more from the use of ChatGPT. On the one hand, institutional investors could benefit more from the use of ChatGPT because prior research suggests that institutional investors are more likely to trade on information than retail investors (e.g., Barber and Odean, 2008; Ben-Rephael et al., 2017). In addition, institutional investors are more familiar with GenAI tools than retail investors. Consistent with this argument, Guo et al. (2024) provide field evidence that more experienced investors benefit more from the use of GenAI-based investment consulting tools. On the other hand, retail investors may benefit more from the use of ChatGPT because compared with institutional investors, retail investors had limited access to information sources and were less equipped to process information before the launch of ChatGPT. For example, according to a survey of 1000 U.S. retail investors, 46 % of those surveyed believe that tools like ChatGPT are useful for picking investments because such tools can help them save time on research.³¹ In addition, several studies document that in other professional settings, less resourceful or skillful workers benefit more from the use of GenAI tools (Brynjolfsson et al., 2023; Noy and Zhang, 2023; Peng et al., 2023).

²⁹ The coefficient on *News* is subsumed after we control for the interaction terms of *News* with fixed effects.

³⁰ The significant coefficient on *Outage* could reflect the possibility that investors also use ChatGPT to facilitate the processing of market- or industry-wide news or unidentified firm-specific news, which is not captured by the *News* variable. It is also consistent with the task complementarity effect discussed in Acemoglu (2024)—ChatGPT outages affect investors' efficiency in conducting other routine tasks, even in the absence of any news, thereby leaving them with less time for trading.

³¹ See <https://www.eto.com/en-us/news-and-analysis/latest-news/press-release/four-in-ten-u-s-retail-investors-use-or-are-open-to-using-chat-gpt-style-ai-to-help-pick-and-manage-investments/>.

Table 4

ChatGPT Outages and Change in Trading Volume: Robustness Tests

This table presents the robustness tests of the impact of ChatGPT outages on trading volume. The baseline regression refers to the specification in Column (2) of Table 3. For brevity, we only report the coefficients on *Outage*, except in Panel F, where we report the coefficient on *Pseudo-Outage*. Panel A reports the Poisson regression results using two alternative measures of trading volume (*Shares Count* and *Trades Count*). Panel B reports the ordinary least squares regression results when controlling for alternative fixed effects. Panel C reports the results using alternative control periods. Panel D reports the results using a median regression or using subsamples. For the median regression, we follow Machado and Silva (2019) to implement the jackknife correction with standard errors clustered by firm and date. Panel E reports the results using OpenAI's start time of investigation as the start time of each outage. Panel F presents the falsification test of the impact of ChatGPT outages on trading volume, using day *t-1* as the pseudo-outage date and days *t-2* through day *t-6* as the corresponding control days. See Appendix C for variable definitions. Standard errors are clustered at the firm and day levels. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively, based on two-sided tests.

	(1)	(2)
	Coeff. on <i>Outage</i>	t/z-statistic
Baseline results (Table 3, Column (2))	−0.084***	−3.25
Panel A: Alternative dependent variables		
<i>Shares Count</i> (Poisson regression)	−0.082***	−2.90
<i>Trades Count</i> (Poisson regression)	−0.073***	−2.78
Panel B: Alternative fixed effects		
Outage × Firm FE and Outage × Time Interval FE	−0.096***	−3.23
Outage × Firm × Time Interval FE	−0.096***	−3.23
Day of Week × Firm FE and Day of Week × Time Interval FE	−0.084**	−2.12
Panel C: Alternative control periods		
Trading days [−10, −1]	−0.069***	−2.96
Trading days [−15, −1]	−0.071***	−3.02
Trading days [−10, −6]	−0.053*	−1.92
Trading days [−15, −6]	−0.059**	−2.18
Panel D: Alternative model or using subsamples		
Median regression	−0.071***	−11.51
Excluding observations with residuals at the 1 % tails	−0.055**	−2.56
Excluding observations with residuals at the 2.5 % tails	−0.039**	−2.18
Excluding the shortest outage	−0.084***	−3.27
Excluding the two shortest outages	−0.087***	−3.74
Excluding the three shortest outages	−0.088***	−3.47
Panel E: Alternative outage start time		
Using OpenAI's start time of investigation	−0.087***	−3.23
Panel F: Pseudo outages		
Using same periods of the previous trading day as pseudo-outage periods	−0.019	−0.34

To test the potential differential effect of ChatGPT outages on retail and institutional trading, we use the method developed by Boehmer et al. (2021) to identify retail investor trading (*Retail Volume*) and use the remaining trading (*Non-Retail Volume*) to proxy for institutional investor trading.³² Columns (1) and (2) of Table 6, Panel A report the regression results for *Retail Volume* and *Non-Retail Volume*, respectively. The coefficient on *Outage* in Column (1) is significantly negative (−0.005; $t = -3.62$), implying a relative decrease of 2.9 % of the sample standard deviation of *Retail Volume*.³³ The coefficient on *Outage* in Column (2) is also significantly

³² Boehmer et al. (2021) identify retail investor trading by exploiting two institutional features of retail trading: (1) most retail trading takes place off-exchange, and (2) retail trading is typically associated with a price improvement corresponding to a small fraction of one cent over the National Best Bid or Offer, with a price marginally below or above a round penny.

³³ While earlier studies suggest that retail investors mostly engage in attention-induced trading (e.g., Barber and Odean, 2008), more recent studies suggest that retail trade imbalance contains information (e.g., Kaniel et al., 2012; Boehmer et al., 2021). Furthermore, Blankespoor et al. (2024) provide survey evidence that nearly half of surveyed retail investors use GenAI tools to process financial information and make investment decisions. Such survey evidence is consistent with our empirical result that ChatGPT outages could affect retail investors' trading decisions.

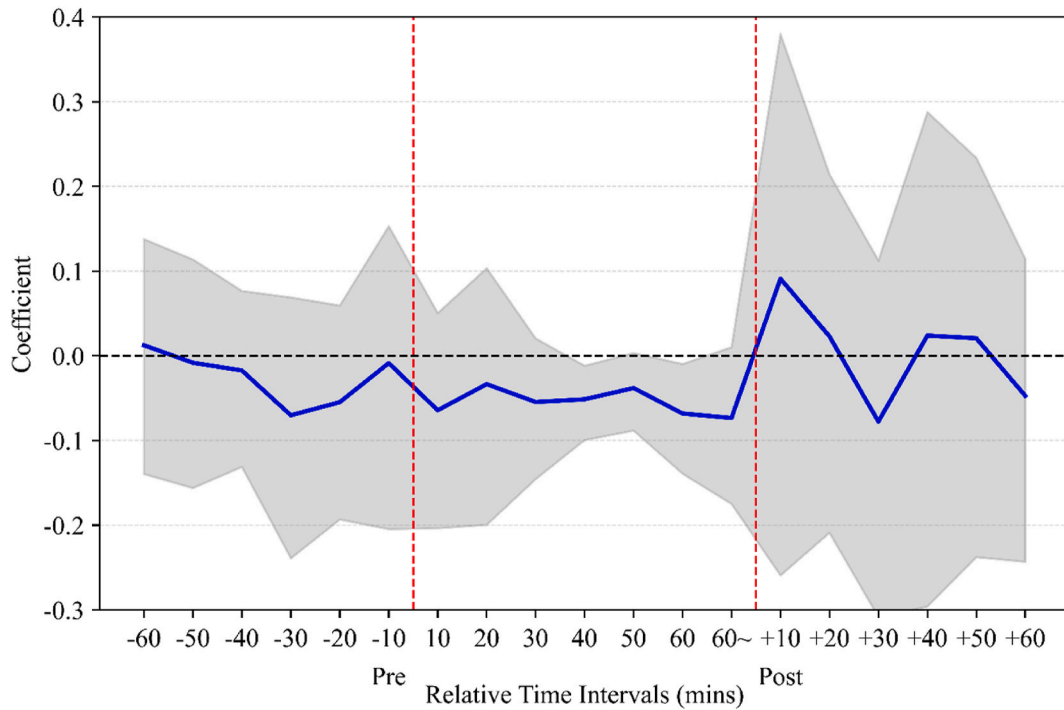


Fig. 1. Changes in Trading Volume Surrounding ChatGPT Outages

This figure illustrates changes in trading volume in 10-min windows surrounding ChatGPT outages. We focus on the three outages that begin after 10:00 a.m. and end before 3:30 p.m. to ensure sufficient pre- and post-periods within trading hours of the same day. The solid line represents the estimated changes in trading volume, which are the coefficients on the interaction terms between *Outage* and the indicators for every 10-min window around outage periods from the regression of trading volume on the interaction terms between *Outage* and the time interval indicators following the same research design as Equation (1), with the shaded area indicating 90 % confidence intervals. The left and right vertical dash lines mark the start and end times of the outages, respectively.

Table 5

ChatGPT Outages and Change in Trading Volume: Conditional on Corporate News Release

This table reports results from the cross-sectional analyses of the impact of ChatGPT outages on trading volume conditional on corporate news release (*News*). The sample includes 2,615,434 security-date-5-minute interval observations from 2553 stock securities during the period from February to August 2023. See Appendix C for variable definitions. Standard errors are clustered at the firm and day level. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively, based on two-sided tests.

Dependent Variable =	Trading Volume	
	(1)	(2)
<i>Outage</i> × <i>News</i>	−0.080* (−1.72)	−0.091** (−2.54)
<i>Outage</i>	−0.067** (−2.64)	−0.058* (−1.95)
<i>News</i>	0.297*** (9.15)	
Controls	YES	YES
Firm FE	YES	YES
Time Interval FE	YES	YES
Controls × <i>News</i>	NO	YES
Firm FE × <i>News</i>	NO	YES
Time Interval FE × <i>News</i>	NO	YES
N	2,615,434	2,615,434
Adjusted R ²	0.458	0.480

negative (-0.079 ; $t = -3.21$), implying a relative decrease of 5.7 % of the sample standard deviation of *Non-Retail Volume*. We further test the difference between these two standardized effects in a stacked regression and find that the effect is significantly larger for non-retail trading than for retail trading ($p = 0.036$).³⁴ However, these results should be interpreted with caution because the approach used by Boehmer et al. (2021) might identify retail trading with a systematic bias (Barber et al., 2024).

Overall, the results indicate that both retail and non-retail investors adopt ChatGPT for trading purposes and that the drop in trading volume is more pronounced for non-retail investors than for retail investors.

4.4.2. Different types of institutional investors

Given the above findings, we conduct cross-sectional analyses to further understand which type of institutional investors' trading is most likely to be impacted by ChatGPT outages. Because the TAQ database does not provide data to link individual trades to specific traders, we rely on the cross-sectional variation in firms' investor bases to examine the heterogeneous effects of ChatGPT outages. For each firm, we obtain institutional ownership data for the most recent quarter from the 13F holdings database. Following Bushee (1998), we classify institutional investors into three categories: transient, dedicated, and quasi-indexer investors. We then calculate the ownership by each type of institutional investors for each firm and create three indicator variables (*Transient_H*, *Dedicated_H*, and *Indexer_H*) based on whether the ownership of each type of institutional investors is above the corresponding sample median for each outage event.

Panel B of Table 6 reports the regression results after adding the three indicator variables and their interactions with control variables and fixed effects. Columns (1) to (3) report the effects for the three types of institutional investors separately. We find that the coefficient on *Outage* $\times$ *Transient_H* is significantly negative ($t = -3.48$), but those on *Outage* $\times$ *Dedicated_H* and *Outage* $\times$ *Indexer_H* are insignificant. In Column (4), we include all three interaction terms in the same regression and obtain the same inferences.

These results indicate that firms with higher ownership by transient institutional investors experience a larger decline in trading volume, suggesting that transient institutional investors are likely to rely more heavily on ChatGPT for real-time information processing and fast trading decisions. In contrast, ChatGPT outages do not have an incremental effect on dedicated institutional investors and indexers. These findings are consistent with insights from our discussions with executives at two fund firms. Executives from the first firm, a long-term dedicated investor, shared that while their employees use ChatGPT for information processing and report drafting, its outages have a minimal impact on their trading decisions, because their trading decisions typically require a long period to formulate and execute. In contrast, executives from the second firm, which offers a trading platform for both individual and institutional investors, shared that the use of ChatGPT greatly improves their efficiency in processing information, including earnings releases and financial statements, enabling investors to trade quickly.

4.5. OpenAI API outages

Lastly, we examine the impact of the outages of OpenAI's API services. Unlike ChatGPT's interactive interface, OpenAI's APIs can only be accessed via programming code. Furthermore, OpenAI provides multiple versions of API services tailored to different use cases, making it difficult to capture the aggregate effect of API outages, which typically impact only a subset of the services. Nevertheless, we identify seven major API outages from January to August of 2023 and determine the start and end time of each outage based on OpenAI's incident reports. We then investigate the change in trading volume during these outage periods relative to the control periods using the same research design as above.

Table 7 reports the regression results. In Columns (1) and (2), we find a modest decline in trading volume during API outages when using total trading volume as the dependent variable.³⁵ To increase the power of the tests, we then investigate the change in algorithmic trading. Recent studies suggest that odd-lot trades are increasingly associated with algorithmic trading, which usually splits trades into smaller pieces (e.g., O'Hara et al., 2014). As a result, following prior studies, we use trades with fewer than 100 shares to measure odd-lot trading volume (*Odd-Lot Volume*). Columns (3) and (4) report the regression results. We find that the coefficient on *Outage* is significantly negative in both columns ($t = -2.24$ and -2.04 , respectively), indicating a significant decline in algorithmic trading during the outage periods of OpenAI's API services.

5. Short-term trading impacts

The above analyses provide evidence of investors using ChatGPT in ways that influence their trading decisions. However, it remains unclear whether ChatGPT enable investors to make more informed trades. Prior research suggests that even publicly available information (e.g., financial disclosures) requires effort to process, and improvements in processing efficiency can help investors incorporate information into stock prices more quickly (Blankespoor et al., 2020; Goldstein et al., 2023). It is thus important to

³⁴ We first standardize the two dependent variables by their respective sample standard deviations to facilitate the comparison of their relative magnitudes. We then pool the two regressions together and include an indicator variable for the observations with non-retail trading volume as the dependent variable, and its interaction with *Outage*, control variables, and fixed effects. We then estimate the pooled regression and assess the differential effect of *Outage* on the two types of trading volumes based on the estimated coefficient on the interaction term between the indicator variable and *Outage*. This approach is analogous to a Chow test (Chow, 1960).

³⁵ In an untabulated analysis, we find that our results in Table 3 remain similar when we exclude the two ChatGPT outages that coincided with OpenAI API outages.

Table 6

ChatGPT Outages and Trading of Different Types of Investors

This table reports the regression results for trading of different types of investors. Panel A reports results from the analyses of the impact of ChatGPT outages on retail (*Retail Volume*) and non-retail trading volume (*Non-Retail Volume*). It also reports the test of the difference in the coefficient on *Outage* scaled by the sample standard deviation of the corresponding dependent variable in a stacked regression. Panel B reports results of the impact of ChatGPT outages on trading volume conditional on the ownership by different types of institutional investors: transient (*Transient_H*), dedicated (*Dedicated_H*), or quasi-indexer institutional ownership (*Indexer_H*). The sample includes 2,615,434 security-date-5-minute interval observations from 2553 stock securities during the period from February to August 2023. See Appendix C for variable definitions. Standard errors are clustered at the firm and day levels. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively, based on two-sided tests.

Panel A: ChatGPT outages and change in retail versus non-retail trading volume				
Dependent Variable =	<i>Retail Volume</i>		<i>Non-Retail Volume</i>	
	(1)		(2)	
<i>Outage</i>	−0.005*** (−3.62)		−0.079*** (−3.21)	
Controls	YES		YES	
Firm FE	YES		YES	
Time Interval FE	YES		YES	
N	2,615,434		2,615,434	
Adjusted R ²	0.383		0.444	
Coefficient on <i>Outage</i> relative to the sample standard deviation of the dependent variable	−0.029***		−0.057***	
Difference	0.028**			
(p-value)	(0.036)			
Panel B: ChatGPT outages and change in trading volume conditional on types of institutional investors				
Dependent Variable =	<i>Trading Volume</i>			
	(1)	(2)	(3)	(4)
<i>Outage</i> × <i>Transient_H</i>	−0.063*** (−3.48)			−0.053*** (−3.79)
<i>Outage</i> × <i>Dedicated_H</i>		0.000 (0.00)		−0.011 (−0.47)
<i>Outage</i> × <i>Indexer_H</i>			0.035 (0.88)	0.040 (1.05)
<i>Outage</i>	−0.051** (−2.17)	−0.073*** (−3.19)	−0.083*** (−3.07)	−0.052** (−2.21)
Controls	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Time Interval FE	YES	YES	YES	YES
Controls × Cross-sectional Variable(s)	YES	YES	YES	YES
Firm FE × Cross-sectional Variable(s)	YES	YES	YES	YES
Time Interval FE × Cross-sectional Variable(s)	YES	YES	YES	YES
N	2,615,434	2,615,434	2,615,434	2,615,434
Adjusted R ²	0.465	0.469	0.467	0.481

understand the potential effects of ChatGPT on investors' informed trades because it has been criticized for its tendency to provide inaccurate information (Athaluri et al., 2023; Emsley, 2023). It is also possible that ChatGPT only provides stale information that has already been incorporated into stock prices.

To answer this question, we examine the effect of ChatGPT outages on the information content of investor trades. We rely on two commonly used empirical proxies to capture the extent of information content embedded in trades: price impact and return variance. We measure a trade's short-term price impact using the stock price movement in the first 5 min following the trade and then calculate the equal-weighted average across all trades in each 5-min interval (*Price Impact*). As shown by Kyle (1985), price impact increases with the amount of private information incorporated into stock prices and decreases with the amount of noise trading. We measure *Return Variance* as the standard deviation of the trade-based returns in each 5-min interval. Prior research shows that this variable reflects the extent of differential information signals that investors trade on (French and Roll, 1986; Holthausen and Verrecchia, 1990). We use the same model as in Equation (1), except that the dependent variable is *Price Impact* or *Return Variance*. Panel A of Table 8 reports the regression results. We find that the coefficient on *Outage* is significantly negative in both columns ($t = -3.58$ and -1.72 , respectively). The results indicate that price impact and return variance decrease significantly during ChatGPT outage periods, consistent with ChatGPT's role in facilitating informed trading. The economic magnitudes are relatively modest; the decreases in *Price Impact* and *Return Variance* relative to their respective standard deviations are 4.93 % ($=0.010/0.203$) and 2.36 % ($=0.003/0.127$), respectively.

Next, to further shed light on whether professional and non-professional investors use ChatGPT similarly, we examine the degree of information asymmetry among investors using *Bid-Ask Spread*, which is calculated as the equal-weighted average of the difference between the best bid and best ask, scaled by their midpoint, for all quotes in each 5-min interval. Prior research (e.g., Glosten and

Table 7

OpenAI API Outages and Change in Trading Volume

This table reports the regression results of the analysis of the impact of OpenAI API outages on total trading volume (*Trading Volume*) and odd-lot trading volume (*Odd-Lot Volume*). The sample includes 1,456,687 security-date-5-minute interval observations from 2547 stock securities during the period from January to August 2023. See [Appendix C](#) for variable definitions. Standard errors are clustered at the firm and day levels. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively, based on two-sided tests.

Dependent variable =	<i>Trading Volume</i>		<i>Odd-Lot Volume</i>	
	(1)	(2)	(3)	(4)
<i>Outage</i>	−0.028* (−2.01)	−0.020 (−1.53)	−0.006** (−2.24)	−0.006** (−2.04)
<i>Return Overnight</i>		2.302*** (3.01)		0.353** (2.52)
<i>Return [−1]</i>		1.163*** (3.89)		0.120* (1.80)
<i>Return [−5, −2]</i>		0.755*** (4.70)		0.091*** (3.53)
<i>Price Range</i>		0.813*** (5.30)		0.173*** (5.84)
<i>Price</i>		0.312 (1.44)		0.175*** (3.73)
<i>Market Cap</i>		−0.591** (−2.63)		−0.139*** (−2.77)
Firm FE	YES	YES	YES	YES
Time Interval FE	YES	YES	YES	YES
N	1,456,687	1,456,687	1,456,687	1,456,687
Adjusted R ²	0.447	0.451	0.525	0.528

[Milgrom, 1985](#)) shows that bid-ask spreads reflect the degree of information asymmetry between informed and uninformed traders. If a ChatGPT outage disrupts investors' informed trading activities to a greater extent for professional investors over retail investors, we would expect a decrease in information asymmetry during the outage periods compared with the control periods. Panel B of [Table 8](#) presents the results. We find that the coefficient on *Outage* is significantly negative ($t = -1.76$), indicating a decrease in information asymmetry during ChatGPT outage periods. The decrease in *Bid-Ask Spread* is modest at 1.68 % ($=0.013/0.776$) of its standard deviations.³⁶

Overall, the modest effects observed in both tests suggest that ChatGPT adoption was either not yet widespread or offered limited advantages to users during our sample period.

6. Long-term price informativeness

The above analyses of short-term trade impacts provide evidence on the positive effect of ChatGPT on investors' informed trading. However, these analyses do not speak to whether ChatGPT-assisted trading affects long-run stock price informativeness. The short-term effects may simply reflect adjustment frictions during ChatGPT outages—such as difficulties in quickly switching to pre-existing tools—rather than a clear comparison of ChatGPT over other tools. If pre-existing tools, such as traditional search engines or in-house information processing technologies, can perfectly substitute for ChatGPT, the short-run effects may dissipate once the adjustment frictions are resolved. Therefore, any long-run effect of ChatGPT on informed trading depends on its relative advantages over pre-existing tools. In this section, we complement the short-term price impact tests by examining the impact of ChatGPT usage on long-run stock price informativeness. Specifically, we construct two firm-level ChatGPT-assisted trading exposure measures as the treatment variable and conduct generalized difference-in-differences (DID) tests to examine the effect of the launch of ChatGPT—a shock to investors' adoption of GenAI tools—on firms' long-term price informativeness. We adopt the price informativeness measure developed by [Bai et al. \(2016\)](#) and [Kacperczyk et al. \(2021\)](#). This measure captures the extent to which current stock prices predict future earnings.

6.1. Construction of ChatGPT-assisted trading exposure measures

We leverage the launch of ChatGPT as an exogenous shock to the availability of GenAI tools for investors. However, this tool became available to all investors of all firms following its debut; therefore, we do not know how the intensity of ChatGPT-assisted trading varies across firms ex ante. To overcome this empirical challenge, we utilize the observed ex-post change in trading volume during ChatGPT outages to quantify ChatGPT-assisted trading exposure (*Exp*), which captures the extent to which a firm's stock

³⁶ Using a sample from 1993 to 2012, [Amiram et al. \(2016\)](#) document a 5.1 % increase in bid-ask spread (relative to the sample mean) on the day of earnings announcements. In comparison, our estimated decrease in bid-ask spread during ChatGPT outages, 3.05 % ($=0.013/0.426$) of the sample mean, is more modest.

Table 8
ChatGPT Outages and Short-Term Trading Impacts

This table reports the regression results for short-term impacts of trades. Panel A reports results from the analyses of the impact of ChatGPT outages on price impact (*Price Impact*) and return variance (*Return Variance*). Panel B reports results from the analyses of the impact of ChatGPT outages on information asymmetry (*Bid-Ask Spread*). The sample includes security-date-5-minute interval observations with required data from 2553 stock securities during the period from February to August 2023. See [Appendix C](#) for variable definitions. Standard errors are clustered at the firm and day levels. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively, based on two-sided tests.

Panel A: ChatGPT outages and change in information content of trades		
Dependent Variable =	<i>Price Impact</i>	<i>Return Variance</i>
	(1)	(2)
<i>Outage</i>	−0.010*** (−3.58)	−0.003* (−1.72)
<i>Return Overnight</i>	−0.098 (−1.29)	−0.080*** (−3.68)
<i>Return [−1]</i>	−0.105* (−1.82)	−0.033** (−2.22)
<i>Return [−5, −2]</i>	−0.056** (−2.47)	−0.041*** (−3.61)
<i>Price Range</i>	0.026*** (3.42)	−0.002 (−0.34)
<i>Price</i>	0.009 (0.97)	0.030*** (2.97)
<i>Market Cap</i>	−0.029*** (−3.55)	−0.051*** (−5.08)
Firm FE	YES	YES
Time Interval FE	YES	YES
N	2,408,754	2,169,531
Adjusted R ²	0.072	0.604
Panel B: ChatGPT outages and change in information asymmetry		
Dependent Variable =	<i>Bid-Ask Spread</i>	
	(1)	
<i>Outage</i>	−0.013* (−1.76)	
<i>Return Overnight</i>	−0.454*** (−5.04)	
<i>Return [−1]</i>	−0.183** (−2.40)	
<i>Return [−5, −2]</i>	−0.232*** (−3.91)	
<i>Price Range</i>	−0.064* (−1.87)	
<i>Price</i>	0.079 (1.23)	
<i>Market Cap</i>	−0.246*** (−3.76)	
Firm FE	YES	
Time Interval FE	YES	
N	2,386,031	
Adjusted R ²	0.741	

trading is affected by ChatGPT.

Specifically, for each outage event, we calculate the change in trading volume for each firm from the control period to the outage period. We multiply this change by −1 so that a higher value indicates a greater decline in trading volume during outage periods. We then take the average of these values over the eight outage events to reduce noise in the measure and define this average value as the firm's ChatGPT-assisted trading exposure (*Exp_cont*). To address heterogeneity in trading volume changes across outages, we rank the drop in trading volume for all firms by each outage into deciles and then take the average of the deciles across the eight outages (*Exp_rank*). Higher values of these two measures imply a larger decline in trading volume during the outage periods and thus greater ChatGPT-assisted trading exposure.

6.2. Determinants of ChatGPT-assisted trading exposure

Before examining the impact of ChatGPT-assisted trading exposure on long-term stock price informativeness, we conduct a

determinant analysis to understand the factors that affect a firm's ChatGPT-assisted trading exposure. We focus on three types of firm characteristics. First, we include variables related to a firm's investor base and information environment, namely institutional ownership (*Inst_Ownership*) and analyst coverage (*Analyst*). Second, we consider information processing difficulties arising from business complexity, namely asset tangibility (*Tangibility*), return volatility (*Ret_Volatility*), R&D intensity (*R&D*), advertising intensity (*Advertising*), and the number of business segments (*Segments*). Third, we include firm fundamentals: firm size (*Size*), an indicator for loss (*Loss*), leverage (*Lev*), and the book-to-market ratio (*B/M*). We measure these variables in 2022, the year before the first ChatGPT outage event in our sample.

Panel A of [Table 9](#) presents the descriptive statistics of the two exposure measures and the determinant variables. Consistent with the results presented in [Section 4](#), the average of the continuous exposure measure (*Exp_cont*) is positive (0.009), indicating an average decrease in trading volume during the outage period.

Panel B of [Table 9](#) reports the regression results. First, we find that firms with higher institutional ownership and analyst coverage have greater ChatGPT-assisted trading exposure, suggesting that a firm's investor base and information environment can explain its ChatGPT-assisted trading exposure. These results are consistent with our previous findings that ChatGPT facilitates investors' information processing activities, especially for institutional investors. Second, we find that firms with less tangible assets, higher stock return volatility, and greater advertising intensity have greater ChatGPT-assisted trading exposure than other firms, indicating that

Table 9**Determinants of ChatGPT-assisted Trading Exposure Measures**

This table reports results from the determinant analysis of firms' ChatGPT-assisted trading exposure (*Exp_cont* and *Exp_rank*). Panel A reports the descriptive statistics of the variables used in the determinant analysis. Panel B reports the regression results. The sample includes 2100 firm-level observations. See [Appendix C](#) for variable definitions. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively, based on two-sided tests.

Panel A: Descriptive statistics						
Variable	N	Mean	SD	p25	p50	p75
<i>Exp_cont</i>	2100	0.009	0.021	−0.001	0.007	0.018
<i>Exp_rank</i>	2100	5.507	1.130	4.625	5.500	6.250
<i>Inst_Ownership</i>	2100	0.729	0.251	0.595	0.812	0.920
<i>Analyst</i>	2100	2.001	0.940	1.386	2.079	2.708
<i>Tangibility</i>	2100	0.201	0.229	0.024	0.110	0.284
<i>Ret_Volatility</i>	2100	0.029	0.013	0.020	0.026	0.036
<i>R&D</i>	2100	0.085	0.183	0.000	0.000	0.066
<i>Advertising</i>	2100	0.015	0.032	0.000	0.000	0.018
<i>Segments</i>	2100	1.109	0.465	0.693	1.099	1.386
<i>Size</i>	2100	14.851	1.855	13.591	14.753	16.041
<i>Loss</i>	2100	0.226	0.418	0.000	0.000	0.000
<i>Lev</i>	2100	0.613	0.271	0.420	0.621	0.847
<i>B/M</i>	2100	0.547	0.422	0.234	0.473	0.793
Panel B: Determinant analyses						
Dependent Variable =	<i>Exp_cont</i>		<i>Exp_rank</i>			
	(1)		(2)			
<i>Inst_Ownership</i>	0.007*** (3.46)		1.011*** (9.09)			
<i>Analyst</i>	0.002** (2.30)		0.131*** (3.19)			
<i>Tangibility</i>	−0.006*** (−3.01)		−0.290*** (−2.66)			
<i>Ret_Volatility</i>	0.236*** (4.90)		20.034*** (7.90)			
<i>R&D</i>	0.003 (0.88)		−0.118 (−0.67)			
<i>Advertising</i>	0.047*** (3.30)		2.222*** (2.95)			
<i>Segments</i>	−0.001 (−0.60)		−0.037 (−0.70)			
<i>Size</i>	−0.000 (−0.72)		−0.083*** (−3.76)			
<i>Loss</i>	0.001 (0.95)		−0.002 (−0.03)			
<i>Lev</i>	0.010*** (4.96)		0.656*** (6.15)			
<i>B/M</i>	0.004*** (3.37)		0.332*** (5.11)			
N	2100		2100			
Adjusted R ²	0.057		0.119			

ChatGPT is especially beneficial for analyzing firms with complex business activities. Lastly, we find that firms with higher financial leverage and book-to-market ratios tend to have greater ChatGPT-assisted trading exposure than other firms, suggesting that contrarian investors rely more on ChatGPT than other investors. These findings illustrate that firm-level characteristics explain investors' tendency to use GenAI tools for trading purposes.

6.3. ChatGPT and stock price informativeness

In this section, we examine how the release of ChatGPT affects long-term stock price informativeness. The sample for this analysis consists of firms for which we can obtain data for the three fiscal quarters before and the three fiscal quarters after the release of ChatGPT in November 2022. We exclude the fiscal quarter surrounding the release of ChatGPT for cleaner identification. Building on the model developed by Kacperczyk et al. (2021), we conduct a generalized DID analysis using the following regression model:

$$\begin{aligned} EBIT/A_{i,q+1} = & \beta_0 + \beta_1 M/A_{i,q} \times Exp_i \times Post_q + \beta_2 M/A_{i,q} \times Exp_i + \beta_3 M/A_{i,q} \times Post_q + \beta_4 Exp_i \times Post_q + \beta_5 M/A_{i,q} + \delta M/A_{i,q} \\ & \times Controls_{i,q} + \gamma Controls_{i,q} + \theta_i + \eta_q + \varepsilon_{i,q+1}, \end{aligned} \quad (2)$$

where i and q indicate firm and fiscal year-quarter, respectively. The earnings variable ($EBIT/A_{i,q+1}$) is defined as the earnings before interest and taxes for firm i in fiscal quarter $q+1$, scaled by total assets at the beginning of $q+1$. The market value variable ($M/A_{i,q}$) is defined as the natural logarithm of the market capitalization of equity scaled by total assets of firm i at the end of fiscal quarter q . $Post_q$ is an indicator variable for the post-ChatGPT period, which includes fiscal quarters that begin after the release of ChatGPT. Exp_i is the treatment variable that captures the extent of firm i 's ChatGPT-assisted trading exposure and is either Exp_cont or Exp_rank . The coefficient of interest is that on $M/A_{i,q} \times Exp_i \times Post_q$. If ChatGPT improves investors' informed trading, we would expect stock prices to incorporate more information about future earnings for firms with greater ChatGPT trading exposure after the release of ChatGPT, and thus we should obtain a positive coefficient on this triple interaction term (β_1).

Following Kacperczyk et al. (2021), we include the following control variables (*Controls*) for firm i in fiscal quarter q : earnings before interest and taxes ($EBIT/A$), firm size (*Size*), leverage (*Lev*), asset tangibility (*Tangibility*), an indicator for foreign business (*Foreign*), cash holdings (*Cash*), institutional ownership (*Inst_Ownership*), and analyst following (*Analyst*). We further control for the interaction terms of $M/A_{i,q}$ with *Exp* and *Post*. We also include interaction terms of *Controls* with $M/A_{i,q}$ to account for the effects of control variables on the relationship between $EBIT/A_{i,q+1}$ and $M/A_{i,q}$. To facilitate result interpretation, we demean *Exp* and control variables in the regression; the coefficient on $M/A_{i,q}$ thus captures the association between $M/A_{i,q}$ and $EBIT/A_{i,q+1}$ when those variables are at their respective sample means. Lastly, we include firm fixed effects (θ_i) to control for time-invariant firm characteristics and fiscal year-quarter fixed effects (η_q) to control for time-trend. The t -statistics are based on standard errors clustered at the firm level.

Table 10 presents the regression results, Column (1) based on *Exp_cont* and Column (2) based on *Exp_rank*. The coefficient on M/A is significantly positive in both columns, consistent with the notion that stock prices incorporate information about future earnings. The coefficient on $M/A \times Exp \times Post$ is significantly positive in both columns ($t = 1.97$ and 2.74 , respectively), suggesting that firms with greater ChatGPT-assisted trading exposure experience an increase in price informativeness after the release of ChatGPT.³⁷ In terms of economic significance, the magnitude of the coefficient in the second column implies that a one-standard-deviation increase in a stock's ChatGPT-assisted trading exposure is associated with a 6.97 % ($= 0.005 \times 1.130/0.081$) increase relative to the average extent to which its stock prices incorporate information about future earnings following the release of ChatGPT. In an untabulated analysis, we examine the pre-ChatGPT trends by estimating the effects separately for quarters $t-3$ and $t-2$, while using quarter $t-1$ as the benchmark. We do not find any significant effects in the pre-ChatGPT periods, supporting the parallel trend assumption.

The results provide early evidence consistent with the notion that GenAI-assisted trading improves a firm's stock price informativeness in the long run. The quasi-randomness of the exact timing of ChatGPT's release allows us to attribute the increase in stock price informativeness to the availability of this GenAI tool. However, due to the selection issue associated with a firm's GenAI-assisted trading intensity, these results may not be generalizable to firms for which investors have chosen not to rely heavily on GenAI tools. In addition, we can only capture a stock's GenAI-assisted trading intensity after the release of ChatGPT, which could be endogenously determined by investors' information processing needs and the stock's information environment. Lastly, given the rapid evolution of GenAI tools and their expanding use cases among investors, their effect could change over time, and thus our results might not be generalizable to future years. As such, our findings should be interpreted with caution.

7. Conclusion

In this paper, we use ChatGPT outages to provide early evidence on the impact of the use of GenAI by investors on their trading decisions and stock price informativeness. We find a significant drop in trading volume during ChatGPT outage periods, indicating that investors use ChatGPT in ways that influence their trading decisions. We further observe more pronounced effects for investors with greater information processing needs and for non-retail investors, particularly transient institutional investors. We also observe a decrease in the degree of information asymmetry among investors during ChatGPT outage periods. Lastly, we provide suggestive

³⁷ We conduct a future earnings response coefficient test, as in Zhu (2019), using a specification based on quarterly earnings—earnings in the current and the next quarter ($Earn_{t+1}$) and their respective four-quarter lags. We do not find a significant coefficient on $Exp \times Post \times Earn_{t+1}$, likely due to the multicollinearity issue in the model.

Table 10
ChatGPT Rollout and Stock Price Informativeness

This table reports results from the analysis of the impact of a firm's ChatGPT-assisted trading exposure on its stock price informativeness using the following model:

$EBIT/A_{i,q+1} = \beta_0 + \beta_1 M/A_{i,q} \times Exp_i \times Post_q + \beta_2 M/A_{i,q} \times Exp_i + \beta_3 M/A_{i,q} \times Post_q + \beta_4 Exp_i \times Post_q + \beta_5 M/A_{i,q} \times \delta M/A_{i,q} \times Controls_{i,q} + \gamma Controls_{i,q} + \theta_i + \eta_q + \varepsilon_{i,q+1}$ The sample includes 12,210 firm-quarter observations from 2060 firms in the three fiscal quarters before and three fiscal quarters after the rollout of ChatGPT in November 2022 (excluding the fiscal quarter surrounding the rollout of ChatGPT). All control variables and Exp are demeaned to facilitate coefficient interpretation. See Appendix C for variable definitions. Standard errors are clustered at the firm level. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively, based on two-sided tests.

Dependent Variable =	$EBIT/A_{q+1}$	
Exp =	Exp_cont	Exp_rank
	(1)	(2)
$M/A \times Exp \times Post$	0.236** (1.97)	0.005*** (2.74)
$M/A \times Exp$	-0.414 (-0.80)	-0.004 (-0.51)
$M/A \times Post$	-0.005* (-1.82)	-0.004* (-1.66)
$Exp \times Post$	0.543*** (2.73)	0.003 (0.88)
M/A	0.085*** (7.73)	0.081*** (7.44)
$EBIT/A$	0.208*** (5.80)	0.208*** (5.80)
$Size$	0.031 (1.14)	0.028 (1.02)
Lev	0.522*** (7.58)	0.523*** (7.44)
$Tangibility$	-0.234* (-1.75)	-0.226* (-1.70)
$Foreign$	-0.016 (-1.24)	-0.016 (-1.27)
$Cash$	-0.035 (-0.75)	-0.035 (-0.76)
$Inst_Ownership$	0.066 (0.97)	0.068 (0.98)
$Analyst$	-0.010* (-1.85)	-0.010** (-1.96)
$M/A \times EBIT/A$	0.130*** (5.54)	0.130*** (5.54)
$M/A \times Size$	-0.004 (-0.53)	-0.004 (-0.63)
$M/A \times Lev$	-0.100** (-2.26)	-0.102** (-2.29)
$M/A \times Tangibility$	0.137*** (3.73)	0.143*** (3.90)
$M/A \times Foreign$	0.008 (0.72)	0.009 (0.83)
$M/A \times Cash$	0.113*** (3.61)	0.112*** (3.47)
$M/A \times Inst_Ownership$	0.021 (0.55)	0.018 (0.46)
$M/A \times Analyst$	0.005 (0.99)	0.005 (0.98)
Firm FE	YES	YES
Fiscal Year-Quarter FE	YES	YES
N	12,210	12,210
Adjusted R ²	0.868	0.868

evidence that ChatGPT facilitates informed trading and improves long-run stock price informativeness.

Although our evidence is suggestive, our study demonstrates the real-world impact of GenAI applications on investor trading shortly after the release of ChatGPT. Our findings indicate that a significant number of investors use ChatGPT in ways that influence their trading decisions, and that GenAI-assisted trading improves long-run stock price informativeness. Future research can investigate the mechanisms through which the adoption of GenAI tools affects trading and market outcomes and the potential issues with using

GenAI tools for trading.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Anecdotal Evidence of Investors' Use Cases of ChatGPT

This appendix provides a summary of anecdotal evidence illustrating investors' use cases of ChatGPT

Use Cases	Source (Date)
Panel A: Information Acquisition	
<i>Firm-level information</i>	
Retrieving information about:	
Business models	Finimize (04/07/2024) ³⁸
Risk profile	Finimize (04/07/2024)
Stock performance	Medium (07/04/2023) ³⁹
Financial ratios	The Motley Fool (06/16/2023) ⁴⁰
Corporate governance	Bloomberg (08/14/2024) ⁴¹
ESG credentials	Finimize (04/07/2024)
Compiling takeaways from earnings news	Bloomberg (02/24/2023) ⁴²
Summarizing regulatory filings and research reports	Bloomberg (08/14/2024)
Summarizing conference calls	Finimize (04/07/2024)
Obtaining recent news	The Motley Fool (06/16/2023)
<i>Industry-level information</i>	
Understanding specialized industry concepts and details	Financial NewsNow (06/13/2023) ⁴³
Summarizing industry policy documents	Bloomberg (10/27/2023) ⁴⁴
Identifying stocks related to a specific theme	J.P. Morgan (07/22/2024) ⁴⁵
<i>Macro-level information</i>	
Summarizing economic reports	Medium (09/30/2024) ⁴⁶
Summarizing tax rule changes	VettaFi (03/06/2023) ⁴⁷
Tracking/summarizing FOMC statements and Fed speeches	Bloomberg (04/27/2023) ⁴⁸
Panel B: Information Integration	
Synthesizing multiple-source information to generate reports	Bloomberg (02/24/2023)
Analyzing sentiment of news and conference calls	Medium (09/30/2024)
Analyzing share price trends	Medium (05/16/2024) ⁴⁹
Forecasting future sales based on historical revenue streams	LivePlan (11/22/2024) ⁵⁰
Making trading recommendations based on earnings reports	Financial News Now (06/13/2023)
Predicting monetary policies and generating tradable signals	Bloomberg (04/27/2023)
Drafting investment commentaries and personalized advice	CFA Institute (05/09/2023) ⁵¹
Conducting portfolio analysis	GOBankingRates (08/24/2023) ⁵²
Panel C: Routine Tasks	
Drafting contracts	CFA Institute (05/09/2023)
Translating reports	CFA Institute (05/09/2023)
Reviewing marketing documents for compliance	CFA Institute (05/09/2023)
Reviewing academic papers	Fortune (06/02/2023) ⁵³
Generating and updating figures	Team-GPT (08/26/2024) ⁵⁴
Assisting financial advisors to access internal research	CNBC (09/18/2023) ⁵⁵
Writing and testing code	Forbes (03/22/2023) ⁵⁶

³⁸ <https://finimize.com/content/how-use-chatgpt-analyze-stock>.

³⁹ <https://medium.com/@muhammadbasitseo/beginner/how-to-use-chatgpt-for-investors-unique-prompts-to-utilize-54e4766960b0>.

⁴⁰ <https://www.fool.com/investing/2023/06/16/how-investors-can-use-the-new-chatgpt-upgrade/>.

⁴¹ <https://www.bloomberg.com/news/articles/2024-08-14/hedge-funds-battle-to-turn-chatgpt-from-intern-to-analyst>.

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Appendix B. Major ChatGPT Outage Events This appendix lists the ChatGPT outage events included in our sample. We collect these events and the level of outage severity from the uptime information disclosed by OpenAI. We determine the outage start time based on the number of user reports regarding ChatGPT service disruption from DownDetector provided by Ookla. For the outage events with timespans beyond trading hours (9:30 to 16:00), we replace the start time and end time with 9:30 and 16:00, respectively. Note that we aggregate trading volume into evenly 5-min time intervals throughout the day. Any 5-min interval that overlaps with an outage is included as part of the outage period. For example, the last 5-min interval that overlaps with the first outage is 14:35:00–14:39:59

Date	Outage Severity	Start Time	End Time	Duration (mins)	Number of 5-min Intervals	Day of Week
02/14/2023	Major	13:55	14:37	42	9	Tuesday
02/27/2023	Major&Partial	09:30	13:30	240	48	Monday
03/15/2023	Major&Partial	14:00	14:18	18	4	Wednesday
03/20/2023	Major	12:20	16:00	220	44	Monday
05/24/2023	Major	15:20	16:00	40	8	Wednesday
06/30/2023	Major	13:10	15:20	130	26	Friday
08/29/2023	Major&Partial	09:55	11:16	81	17	Tuesday
08/31/2023	Major	09:30	12:29	179	36	Thursday

Appendix C. Variable Definitions

Variable Name	Definition
Trading Volume Analyses	
<i>Dependent Variables</i>	
<i>Trading Volume</i>	The total number of shares traded in a 5-min interval, divided by the total number of shares outstanding, then multiplied by 10,000.
<i>Shares Count</i>	The total number of shares traded in a 5-min interval.
<i>Trades Count</i>	The total number of trades in a 5-min interval.
<i>Retail Volume</i>	The total number of shares traded by retail investors in a 5-min interval, divided by the total number of shares outstanding, then multiplied by 10,000. Retail trades are identified using the method developed by Boehmer et al. (2021) .
<i>Non-Retail Volume</i>	The difference between <i>Trading Volume</i> and <i>Retail Volume</i> .
<i>Odd-Lot Volume</i>	The number of odd-lot shares traded in a 5-min interval, divided by the total number of shares outstanding, then multiplied by 10,000.
<i>Independent Variable</i>	
<i>Outage</i>	An indicator variable that equals one if the trading interval coincides with a ChatGPT outage period, and zero otherwise.
<i>Pseudo-Outage</i>	An indicator variable that equals one if the trading interval coincides with the ChatGPT outage period on the day before the outage date, and zero otherwise.
<i>Control Variables</i>	
<i>Return Overnight</i>	Stock return measured from the closing price on day $t-1$ to the opening price on day t .
<i>Return [-1]</i>	Stock return on day $t-1$.
<i>Return [-5, -2]</i>	Cumulative stock return from day $t-5$ through day $t-2$.
<i>Price Range</i>	The logarithm of the highest closing price minus the logarithm of the lowest closing price during the previous 60-day period.
<i>Price</i>	The logarithm of one plus the stock price on day $t-1$.
<i>Market Cap</i>	The logarithm of one plus the product of the stock price and the total number of shares outstanding on day $t-1$.
<i>Cross-Sectional Variables</i>	

(continued on next page)

(continued)

Variable Name	Definition
<i>News</i>	An indicator variable that equals one if there is at least one corporate announcement (i.e., earnings announcement, management forecast, press release from RavenPack Press Release Edition, or SEC filing), news article from RavenPack Dow Jones Edition, or analyst forecast about the firm released on the outage day but before the end of the outage period for the observations from outage days, and zero otherwise. It is similarly defined for observations from control days.
<i>Transient_H (Dedicated_H, Indexer_H)</i>	An indicator variable that equals one if the ownership of transient (dedicated, indexer) institutional investors is higher than the sample median by each outage event, and zero otherwise.
Short-Term Trading Impacts Analyses: Dependent Variables	
<i>Price Impact</i>	Equal-weighted average of the price impact of trades in a 5-min interval, then multiplied by 100. For each trade, the price impact is calculated as $2 \times D \times (M_{t+5} - M_t)/M_t$, where D equals 1 for a buy transaction and -1 for a sell transaction, M_t is the prevailing bid-ask mid-point, and M_{t+5} is the bid-ask mid-point 5 min after the trade. We classify each transaction into a buy or sell transaction using Lee and Ready's (1991) approach.
<i>Return Variance</i>	Standard deviation of trade-based returns in a 5-min interval, then multiplied by 100. We require more than five trades for the 5-min interval for the calculation of this variable.
<i>Bid-Ask Spread</i>	Equal-weighted average of the best bid-ask spread scaled by the mid-quote across all quotes in a 5-min interval, then multiplied by 100.
Long-Term Price Informativeness Analyses	
Dependent Variables	
<i>EBIT/A_{q+1}</i>	Earnings before interest and taxes in fiscal quarter $q+1$ scaled by total assets at the beginning of quarter $q+1$.
Independent Variables	
<i>Exp_cont</i>	A continuous measure of the extent of a firm's share trading facilitated by ChatGPT, calculated as the difference in trading volume for each firm between the outage period and the control period, then multiplied by -1 . For each outage event, the control period includes the corresponding time intervals for the same firm over the preceding five trading days. We then take the average of these differences over the eight outage events to create a firm-level time-invariant continuous measure.
<i>Exp_rank</i>	A rank measure of the extent of a firm's share trading facilitated by ChatGPT, calculated as the decile rankings based on the differences in trading volume between the outage period and the control period for all firms by each event. We take an average of the decile rankings over the eight outage events to create a firm-level time-invariant rank measure.
<i>Post</i>	An indicator variable that equals one if the fiscal quarter begins after the rollout of ChatGPT on November 30, 2022, and zero otherwise.
<i>M/A</i>	The natural logarithm of market capitalization divided by total assets.
Control Variables or Determinants	
<i>Inst_Ownership</i>	The number of shares held by institutional investors scaled by the total number of shares outstanding.
<i>Analyst</i>	The natural logarithm of one plus the number of analysts following the firm.
<i>Tangibility</i>	Net property, plant, and equipment scaled by total assets.
<i>Ret_Volatility</i>	The standard deviation of daily stock returns.
<i>R&D</i>	R&D expenditures scaled by total operating expenses.
<i>Advertising</i>	Advertising expenditures scaled by total operating expenses.
<i>Segments</i>	The natural logarithm of one plus the number of business segments.
<i>Size</i>	The natural logarithm of total assets.
<i>Loss</i>	An indicator variable that equals one if the earnings before extraordinary items in the quarter is negative, and zero otherwise.
<i>Lev</i>	The ratio of total liabilities to total assets.
<i>B/M</i>	The ratio of the book value of equity to the market value of equity.
<i>Foreign</i>	An indicator variable that equals one if the firm has a non-zero foreign exchange income or loss, and zero otherwise.
<i>Cash</i>	Cash holdings scaled by total assets.

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